

Isaac Sim 6.0.1 Physics Re-validation -- Review Data Tables

Post-audit corrected; every experiment traceable to log + reproducible

Environment

Isaac Sim 6.0.1 / Isaac Lab 3.0 (v3.0.0-beta2.patch1), Kit Python 3.12

GPU: NVIDIA GeForce RTX 5090 (31 GiB); driver 610.43.02; Warp 1.13.0 (CUDA 12.9)

Container image: yunchien/isaac:devel (base BASE_IMAGE=nvcr.io/nvidia/isaac-sim:6.0.1)

Physics dt: 1/60 s (unless a per-experiment fine-dt control says otherwise)

Reproduce (from the isaac worktree root)

```
W=/home/<user>/work/worktree/<wt>    # container path of this worktree  
  
just setup apply && just run -t devel -d    # start devel container  
  
# then run each per-page command (env PYTHONPATH=$W/framework is only  
# required for URDF-import drivers but is harmless everywhere)
```

Verdict legend (2026-09 adversarial physics-validity audit)

FLAWED headline conclusion was wrong; fixed + re-verified

QUESTIONABLE design/provenance gap; fixed (control added / re-run) or doc corrected

MINOR conclusion holds; wording / number alignment only

SOUND clean

Tally: 1 FLAWED, 7 QUESTIONABLE, 5 MINOR, 1 SOUND -> all addressed.

Qualitative L2/L2.5/L3 hierarchy was always correct; this pass fixed the
quantitative / mechanistic claims the framework depends on.

#212 Single-joint sag vs stiffness (L2.5/L3)

Verdict: QUESTIONABLE -> FIXED (damping control added)

log: test/.prove-A-212-sag.json sha256:97cd5e2ca3e07f6f

k (N/m)	droop (mm)	mg/k pred (mm)	k_eff/k	drift (mm)
5000.0000	19.4006	19.6200	1.011	0.0000
10000.0000	9.5896	9.8100	1.023	0.0000
1.000e+05	0.7911	0.9810	1.240	0.0000
5.000e+05	0.0763	0.1962	2.571	0.0000
1.000e+06	0.0180	0.0981	5.435	0.0000
--- damping control @ k=1.				
factor 0.25 (d=1581.1)	-0.1023	0.0981		0.0000
factor 0.50 (d=3162.3)	-0.0294	0.0981		0.0000
factor 1.00 (d=6324.6)	0.0180	0.0981		0.0000
factor 2.00 (d=12649.1)	0.0501	0.0981		0.0000
spread=0.1524 mm -> dro				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh $W/src/script/exp_l25_sag_sweep.py
--out $W/test/.prove-A-212-sag.json
```

Corrected conclusion:

Damping control (fixed $k=1e6$, sweep damping) shows droop swings $-0.10..+0.05$ mm and even goes NEGATIVE at low damping -- at true steady state ($\text{drift}=0$) droop must equal $\text{mg}/k=0.098$ mm regardless of damping. So the high- k undershoot (k_{eff}/k up to 5.4x) is a TGS implicit-drive spring-stiffening artifact, NOT a stiffness property. ADR-0021 D1a 'no floor / beats prediction' is refuted; the defensible claim is only 'no float32 floor hit in-range'.

#215 Per-link true-kinematic substitution (L2 hold)

Verdict: QUESTIONABLE (crashed) -> FIXED (NameError; re-run clean)

log: test/.prove-B-215-hold.json sha256:3e4c335d2d83135f

body (link class)	max pos err (m)	max ang err (deg)	pusher frames	zero-err
leaf (box)	6.101e-08	4.518e-06	16	True
internal (box)	1.277e-07	4.518e-06	18	True
base (box)	2.404e-07	4.518e-06	40	True
odd (cylinder)	4.610e-07	4.518e-06	25	True
all_zero_error = True				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/exp_l2_kinematic_substitution.py --out $W/test/.prove-B-215-hold.json
```

Corrected conclusion:

NameError (contact_frames[nm]) fixed -> committed driver now produces the table. Kinematic bodies hold to the float32 read-back floor (6e-8..4.6e-7 m) under 10 kg load + contact, vs a dynamic body that would fall 1.36 mm/step -- the kinematic-vs-dynamic distinction is real. The 'ignores contact' half is true-by-definition (softened, not a stress test).

#216 L3 trajectory tracking precision

Verdict: QUESTIONABLE -> DOC CORRECTED (units + inapplicable floor)

log: test/.prove-A-216-tracking.json sha256:9a65b52fddcf7f1a

k (N/m)	steady-state err (mrad)	rms track err (mrad)	note
500.0000	5.364e-04		per-DOF; revolute axis +Z (zero gra
5000.0000	1.788e-04		per-DOF; revolute axis +Z (zero gra
50000.0000	5.960e-05		per-DOF; revolute axis +Z (zero gra
repeatability k=5000.0000		3.5779	3 runs, spread 0.0000 mrad

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh $W/src/script/exp_traj_tracking.py
--out $W/test/.prove-A-216-tracking.json
```

Corrected conclusion:

Driver reports RADIANS/mrad; the acceptance doc mislabeled them 'mm' (an angle as a distance). The joint is revolute about +Z so gravity gives ZERO torque -> ideal steady-state error ~0, NOT mg/k; the doc's 'analytic floor mg/k=1.962 mm' was imported from the prismatic experiment and is inapplicable. Correct metric: angular tracking error in mrad, no gravity floor.

#219 Drive limits: effort / velocity / position

Verdict: QUESTIONABLE (provenance) -> FIXED (re-run matches driver)

log: test/.prove-A-219-limits.json sha256:a1b2e5e190e439d8

sub-test	case	result	held/clamped
effort clamp	maxForce=1.000e+06N (1e+04x	droop=0.7911 mm	held
effort clamp	maxForce=117.7200N (1.2x load)	droop=0.7911 mm	held
effort clamp	maxForce=49.0500N (0.5x load)	droop=2500.0002 mm	STALL@limit
velocity clamp	limit 0.3 vs 1000	peak 70.3319 -> 0.3031 m/s	True
position limit	cmd 2.0000 m, upper 0.5000 m	settled 0.5003 m	True
solver iters	1 / 4 / 32 iters	droop spread 0.2378 mm	iter-indep=True
gain scaling	revolute (angular)	stored/Kp=0.0175	match=True
gain scaling	prismatic (linear)	stored/Kp=1.0000	match=True

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh $W/src/script/exp_drive_limits.py
--out $W/test/.prove-A-219-limits.json
```

Corrected conclusion:

Re-run so numbers match the committed driver. Payload weight is 98.1 N (10 kg) -- the doc's '49 N weight' mislabeled a 0.5x maxForce CASE (49.05 N) as the weight. maxForce $\geq$ mg holds (droop mg/k); maxForce $<$ mg stalls at the lower limit. Position limit 0.5 m clamps a 2.0 m command. Droop is solver-iteration-independent; revolute drive stores $Kp \cdot \pi / 180$.

#218 Kinematic carry speed limit (analytic d_crit)

Verdict: MINOR (report mu-dependent threshold)

log: test/.prove-B-218-carry.json sha256:c4e36fe2c8f68792

mu	max carried step (m)	min slipped step (m)	analytic d_crit (m)	monotonic
0.20	0.0300	0.0400	0.0346	True
0.50	0.0500	0.0600	0.0547	True
1.00	0.0600	0.0750	0.0774	True
friction_monotonic = True				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/exp_l2_carry_speed_limit.py --out $W/test/.prove-B-218-carry.json
```

Corrected conclusion:

Bracket derivation sound: $d_{crit} = dt \cdot \sqrt{2 \cdot \mu \cdot g \cdot L_{back}}$ matches the measured carried/slipped threshold and rises monotonically with μ (0.035..0.077 m/tick). Report the μ -dependent threshold, not a coarse single band; the high-speed 'launch to 5.19 m' endpoint is a contact-penetration artifact, not a physical finding.

#220 Kinematic pushes dynamic (one-way transfer)

Verdict: QUESTIONABLE -> WORDING SOFTENED

log: test/.prove-B-220-push.json sha256:3aaf950bcf08f258

object	metric	value
plate (kinematic)	max pos err under contact (m)	6.001e-08 [float32 floor; true-by-def]
plate (kinematic)	contact frames	182
cube (dynamic)	displacement x (m)	1.1933 [genuine push]
transfer	one_way_transfer_ok	True

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh $W/src/script/exp_l2_push_dynamic.py
--out $W/test/.prove-B-220-push.json
```

Corrected conclusion:

Plate 'tracks 0.000 under contact' is true-by-definition of kinematic (teleport per tick + never integrated -> any reaction discarded before read-back), not a measurement -- stated as such. The genuine finding is the cube-push: normal-overlap depenetration imparts ~1.19 m displacement. One-way transfer holds.

#221 Maximal loop-joint seam compliance

Verdict: QUESTIONABLE -> REFRAMED AS UPPER BOUND

log: test/.prove-C-221-seam.json sha256:1b02a8ad2a8c9a3c

mass (kg)	static give z (m)	body travel (m)	built
1.0000	2.384e-08		
10.0000	2.384e-08		
100.0000	2.384e-08		
1000.0000	2.384e-08		
--- give identical across 1000x mass			
upper bound: give < ~2e-7 m @ 9810 N	=> seam stiffness	> ~5e10 N/m	
compliant_confirmed=False force	chain variant (#308): SIGSEGV, no		

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/exp_l2_loop_joint_boundary.py --out $W/test/.prove-C-221-seam.json
```

Corrected conclusion:

static_give is bit-for-bit identical (2.38e-8 m) across 1/10/100/1000 kg -> it is the float32 quantization residual, not a measured compliance. The run yields only an UPPER BOUND: give < ~2e-7 m at 9810 N => seam stiffness > ~5e10 N/m (effectively rigid for practical loads). Drop the specific give values, the 'mass-invariant' / monotonic sub-claims, and 'refutes PhysX #308' -- #308 is about CHAINED joints; only a single joint ran (the chain variant SIGSEGVs).

#227 3-joint serial sag accumulation

Verdict: MINOR (geometric-identity wording)

log: test/.prove-A-227-multijoint.json sha256:fa8cfe798352a78c

joint	supported mass (kg)	sag (mm)	mg/k pred (mm)	drift (mm)
joint_0	15.0000	30.3389	29.4300	0.0000
joint_1	10.0000	20.0532	19.6200	0.0000
joint_2	5.0000	9.8164	9.8100	0.0000
coupling: moved joint_0 to				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh $W/src/script/exp_multijoint_sag.py
--out $W/test/.prove-A-227-multijoint.json
```

Corrected conclusion:

Clean vertical-prismatic stack; measured tip sum 60.3 mm vs predicted 58.9 mm (+2.4% real coupling).
'tip error = sum of per-joint sags' is a geometric identity of a serial chain, not a discovered solver
coupling -- the genuine finding is measured-sum ~ = predicted-sum.

#229 Base carries a floating articulation (negative)

Verdict: MINOR (doc numbers corrected to committed JSON)

log: test/.prove-C-229-basecarry.json sha256:aabc3db88a881def

lane	base travel (m)	arm travel (m)	follow ratio	carries
usd_child	2.5000	-1.939e-24	-7.755e-25	False
finding: topology A (USD-hi				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/exp_l2_base_carries_arm.py --out $W/test/.prove-C-229-basecarry.json
```

Corrected conclusion:

Genuine PhysX negative: a USD-hierarchy parent does NOT carry a floating articulation (follow_ratio ~0, arm rides 0 while base moves 2.5 m) -- a joint is forced. The acceptance doc's numbers (base 1.5 m / arm diverges to 2.369 m) were STALE and even implied divergence the clean follow=0 run contradicts; corrected to the committed JSON (base 2.5 / arm 0 / follow 0).

D1 L2.5 dynamic push: pusher back-off vs stiffness

Verdict: QUESTIONABLE -> FIXED (no-cube control + feed-forward)

log: test/.l25-dynamic-push.json sha256:4813b0f2161bcd5c

k (N/m)	cube disp (m)	backoff contact (mm)	control lag (mm)	contact reaction (mm)
10000.0000	1.2071	3.0218	3.8874	0.0000
1.000e+05	1.2001	2.5261	2.3979	0.1283
1.000e+06	1.2000	0.8150	0.7999	0.0151
1.000e+07	1.2004	0.2531	0.2531	0.0000
feed-forward=True; contac				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/exp_l25_dynamic_interaction.py --mode push --out $W/test/.l25-dynamic-push.json
```

Corrected conclusion:

Added drive velocity feed-forward + an identical NO-CUBE control lane. Result: control (following-lag) back-off ~ = the contact back-off at every k, so the PURE contact reaction is ~0 (<0.13 mm). The soft-end 'back-off' the original attributed to contact was drive-following / startup lag, not finite-spring reaction -- confirming the audit.

D2 L2.5 dynamic carry: payload lag vs stiffness + dt

Verdict: FLAWED -> FIXED (feed-forward + dt-refinement control)

log: test/.l25-dynamic-carry.json sha256:f164400584876293

k (N/m)	payload lag std-dt (mm)	payload lag fine-dt (mm)	delivery ok (std)	carrier err (mm)
10000.0000	0.0271	-8.318e-04	True	14.0800
1.000e+05	153.8777	18.6998	False	10.1984
1.000e+06	203.5892	151.4822	False	3.7487
1.000e+07	198.1573	243.7198	False	1.1861
std dt=0.0167 ; fine dt=0.004166667				

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/exp_l25_dynamic_interaction.py --mode carry --out $W/test/.l25-dynamic-carry.json
# fine-dt control (4x): --dt 0.004166667 --steps 1200 --warmup 240 --settle 480 --out
$W/test/.l25-dynamic-carry-finedt.json
```

Corrected conclusion:

The 'stiffer is worse for the payload' conclusion was BACKWARDS. First principles: $a=0.75 \text{ m/s}^2$ needs 0.75 N friction vs 4.9 N static limit (6.5x) -> zero slip in continuous motion at any k. The high-k slip is a $(k*dt)$ discretization artifact: a stiff drive snaps the per-tick position staircase impulsively when dt is not $\ll$ the drive natural period $2*\pi*\sqrt{m/k}$. It COLLAPSES with 4x finer dt (k=1e5: 154 mm -> 19 mm; k=1e6/1e7 need finer still). Delivery is judged on the PAYLOAD lag, not the carrier tracking error (which can be $<0.05 \text{ mm}$ while the cargo lags). L2.5 CAN carry; match dt to stiffness.

Limit (1) Articulation + kinematic maximal loop-joint SIGSEGV

Verdict: MINOR (correctly stated; upstream bug)

log: (none -- see command)

control	outcome
dynamic body (no ArticulationRootAPI)	exit 0 (survives)
floating articulation + get_transforms	SIGSEGV
body-only readback	survives
no readback	survives
=> trigger = artic topology x get_transforms on kinematic anchor a	

Reproduce:

```
just exec -t devel env PYTHONPATH=$W/framework /isaac-sim/python.sh
$W/src/script/repro_artic_kinematic_loopjoint_sigsegv.py # native SIGSEGV -> writes no JSON
```

Corrected conclusion:

Clean bisection isolates a native SIGSEGV in libomni.physx.tensors to a documented-legitimate PhysX construct -> upstream engine defect (isaac-sim/IsaacSim#803), workaround = plain dynamic body. Note the older ADR-0008 wording ('crash at physics-view init') is stale; the repro pins it to the post-step get_transforms readback.

Limit (2) PhysX forbids kinematic articulation LINKS

Verdict: MINOR (correctly stated; PhysX 5.4 doc)

log: (none -- see command)

fact	source
links inside an articulation cannot be kinematic	PhysX 5.4 Articulations doc
true-L2 substitution target = standalone rigid body	ADR-0008 clause 4 / ADR-0021 D2
#229 proves the COMPLEMENTARY proposition	supporting, not proof of the clause

Reproduce:

(documentation constraint; #229 `usd_child` is supporting evidence, not proof)

Corrected conclusion:

Core claim correctly attributed to PhysX 5.4 docs. Minor: the doc's '#229 proves this' over-attributes -- #229 shows a floating root is not dragged by a kinematic USD parent, which is supporting evidence, not a proof that links cannot be kinematic.

Limit (3) #205 / #171 / #166 deferred on missing CAD

Verdict: SOUND (genuinely asset-blocked)

log: (none -- see command)

experiment	blocker
#205 real-vehicle e2e	needs user CAD model (ADR-0021)
#171 representative-robot e2e	needs CAD model
#166 DAE color import	needs CAD (pipeline in ADR-0020)

Reproduce:

(no run: blocked on the user's CAD model)

Corrected conclusion:

Confirmed genuinely asset-blocked, not a technical impossibility: the DAE color-import pipeline (ADR-0020) and the CAD dependency (ADR-0021) are specified; only the asset is missing.